

2.19 能量

(1) 不同量对应同一波函数的情况，
最典型的是能量与动量，即能量与动量
又对应能量与动量。

$$\text{以H原子为例: } E_n = -\frac{e^2}{2an}$$

$$\text{其波函数为 } \psi(r, \theta, \phi) = R(r) Y(\theta, \phi)$$

如Lm简并，由 $0 \leq l < n$

$$-l \leq m \leq l$$

$$\text{故简并度为 } \sum_{m=-l}^l (2l+1) = n^2$$

若考虑电子的自旋，又加 $m_s = \pm \frac{1}{2}$ 简并
总简并度为 $2n^2$

$$(2) \quad E_n = E_n^{(0)} + H_n^{(1)} + \sum_{m \neq n} \frac{|H_{nm}^{(1)}|^2}{E_n^{(0)} - E_m^{(0)}}$$

(3) 微扰论中求能量修正的算符

$\hat{H} = \hat{H}_0 + \hat{H}'$ 其中 $\hat{H}'$ 为微扰

$$\hat{H}' |i, l_1\rangle = \lambda |i, l_1\rangle$$

$$\hat{H}' |i, l_2\rangle = \lambda' |i, l_2\rangle$$

$$\hat{H}'^2 |i, l_1\rangle = \lambda^2 |i, l_1\rangle$$

$$\lambda^2 = \pm 1$$

$$\text{当 } \lambda = 1 \text{ 或 } -1$$

$$\text{当 } \lambda = -1 \text{ 或 } 1$$

所以算符 $\hat{H}'$ 的性质，其自伴算符及反对称
为反对称算符，故为反对称算符

(4) 由外场自伴相位的自伴算符而导致的
算符 $\hat{H}'$ 为反对称算符，而由自伴算符
导致的算符为自伴算符

$$V(r) = -V_0 \delta(r), V_0 > 0$$

$$(1) \quad \text{薛定谔方程: } \hat{H} \psi(r) = E \psi(r)$$

$$\text{SE: } i\hbar \frac{\partial}{\partial t} \psi(r, t) = -\frac{\hbar^2}{2m} \nabla^2 \psi(r, t) - V_0 \delta(r) \psi(r, t)$$

$$\text{定态: } -\frac{\hbar^2}{2m} \nabla^2 \psi(r) - V_0 \delta(r) \psi(r) = E \psi(r)$$

(2) 束缚态

$$\therefore E < 0$$

$$\psi(r) = A e^{-\alpha r} \quad r > 0$$

$$B e^{\alpha r} \quad r < 0$$



(2)

$$\psi'(0^+) - \psi'(0^-) = -\frac{2m V_0}{\hbar^2} \psi(0)$$

$$\text{又: } \psi(0^+) = \psi(0^-)$$

$$\therefore A = B$$

$$\therefore A(1) - B(1) = -\frac{2m V_0}{\hbar^2} A$$

$$\therefore \alpha = \frac{m V_0}{\hbar^2}$$

$$\text{又: } \alpha = \sqrt{\frac{2m E}{\hbar^2}}$$

$$\therefore E = -\frac{m^2 V_0^2}{2 \hbar^2}$$

(3)

$$1 = \int_{-\infty}^0 A^2 e^{2\alpha r} dr + \int_0^{\infty} A^2 e^{-2\alpha r} dr$$

$$A = \sqrt{2}$$

$$= \sqrt{\frac{m V_0}{\hbar^2}}$$

$$\psi(r) = \begin{cases} \sqrt{\frac{m V_0}{\hbar^2}} e^{-\alpha r} & r > 0 \\ \sqrt{\frac{m V_0}{\hbar^2}} e^{\alpha r} & r < 0 \end{cases} \quad \alpha = \frac{m V_0}{\hbar^2}$$

$$\text{归一化: } \int_{-\infty}^{\infty} |\psi(r)|^2 dr = 1$$

$$\frac{1}{4} = \int_0^{\infty} 2 e^{-2\alpha r} dr \quad \text{解得 } \alpha = \frac{1}{2}$$

$$\frac{d\langle L_z \rangle}{dt} = \omega$$

$$\text{三. } U(n) = \frac{1}{2} m \omega^2 x^2$$

$$U(n) \langle U \rangle:$$

$$(1) \text{ 在坐标表象中 } \psi(n) = \langle n | \psi \rangle = N_n e^{-\frac{\alpha^2}{2}} U(\alpha x)$$

$$\begin{aligned} \therefore \langle U \rangle &= \langle n | U | n \rangle & \text{又: } \psi(n) &= \frac{1}{\sqrt{2}} \left[\sqrt{\frac{n+1}{2}} \psi(n+1) + \sqrt{\frac{n-1}{2}} \psi(n-1) \right] \\ &= \langle n | \frac{1}{2} m \omega^2 x^2 | n \rangle & \alpha &= \sqrt{\frac{m \omega}{\hbar}} \\ &= \frac{1}{2} m \omega^2 \langle n | x^2 | n \rangle \\ &= \frac{1}{2} m \omega^2 \frac{1}{\alpha^2} \left[\frac{n}{2} + \frac{n+1}{2} \right] \\ &= \frac{1}{2} \hbar \omega \end{aligned}$$

$$(2) \quad \hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2} m \omega^2 x^2 \quad \text{用 Virial 定理, 求期望}$$

$$\langle \frac{\partial H}{\partial x} \rangle = \frac{\partial E_n}{\partial \omega}$$

$$\frac{\partial \langle U \rangle}{\partial \omega} = (n + \frac{1}{2}) \hbar$$

$$\langle U \rangle = \frac{1}{2} (n + \frac{1}{2}) \hbar \omega$$

$$\text{四. } \hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$$

$$(1) \quad [\hat{L}_x, \hat{p}] = i \hbar$$

$$[\hat{L}_x, \hat{L}_y]$$

$$= [\hat{L}_x, y \hat{p}_z - z \hat{p}_y] = [\hat{L}_x, y \hat{p}_z] - [\hat{L}_x, z \hat{p}_y]$$

$$= [\hat{L}_x, y \hat{p}_z] + [\hat{L}_x, z \hat{p}_y] + [\hat{L}_x, y \hat{p}_z] + [\hat{L}_x, z \hat{p}_y]$$

$$= y [\hat{L}_x, \hat{p}_z] + x [\hat{L}_x, \hat{p}_y]$$

$$= i \hbar [x \hat{p}_y - y \hat{p}_x]$$

$$= i \hbar \hat{L}_z$$

$$\text{同理由 } \hat{L}_x, \hat{L}_y, \hat{L}_z = i \hbar \hat{L}_x, \hat{L}_y, \hat{L}_z$$

$$[\hat{L}_x, \hat{L}_x] = i \hbar \hat{L}_y$$

$$(1) \quad \langle \hat{L}_x \rangle = \langle \psi | \hat{L}_x | \psi \rangle$$

$$= m \hbar$$

$$(1) \quad \langle \hat{L}_x \rangle, \langle \hat{L}_y \rangle, \langle \hat{L}_z \rangle$$

$$\text{① } \langle \hat{L}_x \rangle = \langle \psi | \hat{L}_x | \psi \rangle$$

$$\text{由 } \hat{L}_x = \hat{L}_y^2 - \hat{L}_z^2 \quad \langle \hat{L}_x \rangle = \langle \hat{L}_y^2 \rangle$$

$$\langle \hat{L}_x^2 \rangle = \langle \hat{L}_y^2 \rangle + \langle \hat{L}_z^2 \rangle$$

$$\therefore \langle \hat{L}_x^2 \rangle = \langle \hat{L}_y^2 \rangle$$

$$= \langle \psi | \hat{L}_x^2 | \psi \rangle$$

$$= \langle \psi | (\hat{L}_x^2 - \hat{L}_z^2) | \psi \rangle$$

$$\therefore \langle \hat{L}_y^2 \rangle = \langle \hat{L}_x^2 \rangle$$

$$= \frac{L(L+1)\hbar^2 - m^2 \hbar^2}{2}$$

$$\langle \hat{L}_z^2 \rangle = \langle \psi | \hat{L}_z^2 | \psi \rangle$$

$$= m^2 \hbar^2$$

$$\text{② } \langle \hat{L}_x^2 \rangle = \langle \psi | \left(\frac{\hat{L}_x + \hat{L}_z}{2} \right) \left(\frac{\hat{L}_x + \hat{L}_z}{2} \right) | \psi \rangle$$

$$\text{设 } |\psi\rangle = |l, m\rangle = \left| \left(\frac{\hat{L}_x + \hat{L}_z}{2} \right) |l, m\rangle \right|^2$$

$$= \left| \frac{1}{2} [\hat{L}_x + \hat{L}_z] |l, m\rangle \right|^2$$

$$= \frac{1}{4} [\hat{L}_x + \hat{L}_z]^2 |l, m\rangle$$

$$= \frac{L(L+1)\hbar^2 - m^2 \hbar^2}{2}$$

$$\text{五. 证明: } \hat{S}_x = \frac{\hbar}{2} \hat{\sigma}_x$$

$$(1) \quad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$[\sigma_x, \sigma_y] = 2i \sigma_z \quad [\sigma_y, \sigma_z] = 2i \sigma_x$$

$$[\sigma_z, \sigma_x] = 2i \sigma_y$$

$$(2) \quad \sigma_x^2 = \hat{1}, \quad \sigma_y^2 = \hat{1}, \quad \sigma_z^2 = \hat{1}$$

$$\sigma_x = 1, \quad \sigma_y = 1, \quad \sigma_z = 1$$

$$\sigma_x = -1, \quad \sigma_y = -1, \quad \sigma_z = -1$$

目多

$$(3), \vec{\mu} = \gamma \vec{S}, \vec{B} = B \vec{e}_z$$

$$H = -\vec{\mu} \cdot \vec{B}$$

$$= -\gamma \vec{S} \cdot \vec{B}$$

$$= -\gamma B S_z$$

$$= -\gamma B S_z$$

$$\psi(0) = \psi$$

$$= \frac{1}{\sqrt{2}} |1\rangle + \frac{1}{\sqrt{2}} |0\rangle$$

~~$\psi(0)$~~

$$\psi(t) = e^{-iHt/\hbar} \psi(0)$$

$$= \frac{1}{\sqrt{2}} e^{-\frac{i\gamma B t}{2}} |1\rangle + \frac{1}{\sqrt{2}} e^{\frac{i\gamma B t}{2}} |0\rangle$$

答. 然.